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Computational mechanisms underlying the impact of Pavlovian bias on instrumental learning in problematic social media users

LU LIU[1], YI-XU PANG[1], ZHI-HAO SONG[1], SI-JIA CHEN[1], YING-YI HAN[1] and YUAN-WEI YAO[2,3]*

[1] Department of Psychology, Sun Yat-sen University, Guangzhou, China

[2] Department of Psychology, The University of Hong Kong, Hong Kong SAR, China

[3] State Key Laboratory of Brain and Cognitive Sciences, The University of Hong Kong, Hong Kong SAR, China

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FULL-LENGTH REPORT

ABSTRACT

Background and aims: Problematic social media use (PSMU), a potential behavioral addiction, has become a worldwide mental health concern. An imbalanced interaction between Pavlovian and instrumental learning systems has been proposed to be central to addiction. However, it remains unclear whether individuals with PSMU also over-rely on the Pavlovian system when flexible instrumental learning is required. *Methods:* To address this question, we used an orthogonalized go/no-go task that distinguished two axes of behavioral control during associative learning: valence (reward or punishment) and action (approach or avoidance). We compared the learning performance of 33 individuals with PSMU and 32 regular social media users in this task. Moreover, latent cognitive factors involved in this task, such as learning rate and reward sensitivity, were estimated using a computational modeling approach. *Results:* The PSMU group showed worse learning performance when Pavlovian and instrumental systems were incongruent in the reward, but not the punishment, domain. Computational modeling results showed a higher learning rate and lower reward sensitivity in the PSMU group than in the control group. *Conclusions:* This study elucidated the computational mechanisms underlying suboptimal instrumental learning in individuals with PSMU. These findings not only highlight the potential of computational modeling to advance our understanding of PSMU, but also shed new light on the development of effective interventions for this disorder.

KEYWORDS

problematic social media use, Pavlovian bias, instrumental learning, computational modeling, reward sensitivity

INTRODUCTION

Social media use has become a near-ubiquitous part of our daily life over the last decade. As of 2023, there were more than 4.5 billion active social media users (https://www.statista.com/statistics/278414). Despite its popularity and convenience, the negative effects of problematic social media use (PSMU) on the physical health and social functioning have been widely reported (Sun & Zhang, 2021). PSMU has been considered a potential behavioral addiction given its similarities to other established addictions, although its classification is still under debate, partly because of the lack of empirical studies examining the cognitive and neurobiological mechanisms associated with this disorder (Brand et al., 2020; Cataldo, Billieux, Esposito, & Corazza, 2022; Montag, Marciano, Schulz, & Becker, 2023; Wegmann, Müller, Turel, & Brand, 2020).

*Corresponding author.
E-mail: ywyao@hku.hk

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Altered associative learning, such as automatic approach to addictive stimuli, has been hypothesized to play a critical role in PSMU and other addictions (Heinz et al., 2019; Lindström et al., 2021; Liu et al., 2024; Meshi, Elizarova, Bender, & Verdejo-Garcia, 2019; Redish & Johnson, 2007). Associative learning is primarily controlled by the Pavlovian and instrumental systems (Gershman, Guitart-Masip, & Cavanagh, 2021). The Pavlovian system supports the acquisition of intrinsic stimulus-outcome associations, whereas the instrumental system supports a more flexible learning of the optimal response to each stimulus based on related outcomes (Guitart-Masip, Duzel, Dolan, & Dayan, 2014; Queirazza, Steele, Krishnadas, Cavanagh, & Philiastides, 2023). These two systems often generate the same predictions to facilitate efficient learning. However, they may interfere with each other in some cases and lead to sub-optimal choices (Guitart-Masip et al., 2012). These sub-optimalities are usually driven by two interdependent axes of behavioral control: valence (i.e., reward or punishment) and action (i.e., approach or avoidance) (Guitart-Masip et al., 2014). Specifically, the Pavlovian system inherently favors approaching reward-related stimuli and avoiding punishment-related stimuli. The instrumental system, on the other hand, allows flexible action selection based on task rules, such as avoidance to obtain rewards (Queirazza, Steele, Krishnadas, Cavanagh, & Philiastides, 2023). Examining the influence of Pavlovian cues on instrumental behavior can advance our understanding of the formation of maladaptive behaviors (e.g., habitual drug seeking) in addictive disorders (Heinz et al., 2019). However, few experimental studies have directly tested whether and how this process goes awry in PSMU.

A popular paradigm that has been used to examine the effect of Pavlovian conditioning on instrumental behavior is the Pavlovian-to-instrumental transfer (PIT) task. Higher PIT effects suggest a greater impact of the Pavlovian system on instrumental behavior (Garbusow et al., 2022). Although studies using the PIT task have provided valuable information on how Pavlovian system biases instrumental behavior in addictive disorders (Qin et al., 2023; Vogel et al., 2018; Xu et al., 2024), this task separates Pavlovian and instrumental conditioning, and the interaction between the two systems is not tested directly during acquisition but in an extinction stage (Guitart-Masip et al., 2014). This is a crucial omission as the acquisition and extinction may rely on different neural and cognitive mechanisms (Sun et al., 2011). Furthermore, most PIT studies focused on the effects of Pavlovian cues on instrumental approach behavior (a few exceptions include Xu et al. (2024)), which limited a comprehensive investigation of the effects of valence and action on instrumental behavior (Chen et al., 2021). These issues can be addressed by the orthogonalized go/no-go task, in which valence and action are independently manipulated, such that the Pavlovian bias can interfere with or facilitate instrumental learning depending on the congruence of the two systems (Guitart-Masip et al., 2012).

More importantly, the orthogonalized go/no-go task allows for a computational analysis to examine the role of related latent cognitive factors (e.g., learning rate) during instrumental learning (Guitart-Masip et al., 2012). These computational models provide an effective approach for precisely investigating the cognitive processes that give rise to the behavioral characteristics of addiction (Gueguen, Schweitzer, & Konova, 2021; Perales, Flayelle, Verdejo-Garcia, Clark, & Billieux, 2024; Robinson, Chong, & Verdejo-Garcia, 2022). Although a few studies have reported learning and decision-making deficits in individuals with PSMU (Meshi et al., 2019; Müller et al., 2023), the computational modeling approach has been rarely used to explore whether and how different cognitive processes become dysfunctional during associative learning (Kato et al., 2023).

Therefore, we used an orthogonalized go/no-go paradigm to examine how Pavlovian biases influence instrumental learning in individuals with PSMU. We hypothesized that the PSMU group would show worse learning performance when Pavlovian and instrumental systems are incongruent in both reward and punishment domains. Using computational modeling, we further examined the latent cognitive processes underlying the between-group differences in learning performance and their associations with the model-free task performance and self-reported measures.

MATERIAL AND METHODS

Participants

A total of 33 individuals with PSMU and 32 regular social media users were recruited from the local Universities to complete the experiment. PSMU severity was assessed using the Bergen Social Networking Addiction Scale (BSNAS; Andreassen, Pallesen, & Griffiths, 2017). All participants had normal or corrected-to-normal vision, no history of neurological or psychiatric disorders, and were free from current psychoactive medication use. The sample size is similar to previous studies using the orthogonalized go/no-go task and computational modeling (Guitart-Masip et al., 2012; Kim et al., 2023; Moutoussis et al., 2018) and gives us 80% power to detect a large between-group effect (e.g., Cohen's $d = 0.75$). We recruited regular social media users with low PSMU severity as the healthy control (HC) group, since we are particularly interested in the impact of PSMU, but not general time spent on social media, on associative learning. Information related to the internet use, including online video clip and live watching/listening time, online social communicating time, and online gaming time, was also measured. All participants were compensated 30 RMB for their time. Moreover, to motivate participants to perform the task carefully, they were told that they would receive a bonus (0-6 RMB) based on the accumulated points earned on the task.

Measures

The BSNAS was used to measure PSMU severity, and a cut-off threshold of 19 was used to separate the PSMU and HC groups as suggested by previous research (Bányai et al., 2017).

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The BSNAS is a six item self-report assessment of the usage of social media (Andreassen et al., 2017). Participants rated all items on a 5-point Likert scale. Total scores ranged from 6 to 30. The BSNAS has been widely used in previous PSMU studies and has shown good psychometric properties (Andreassen et al., 2017; Bányai et al., 2017).

Moreover, a subset of participants (n = 46, including 27 PSMU and 19 HC participants) completed the Barratt Impulsiveness Scale version 11 (BIS-11; Patton, Stanford, & Barratt, 1995) and the Behavioral Inhibition System/Behavioral Approach System (BIS/BAS; Carver & White, 1994) scales. The BIS-11 consists of 30 items on a 4-point Likert scale. The total score of the BIS-11 was used as an index of self-reported impulsivity. The BIS/BAS scales include 20 items on a 4-point Likert scale. The scores of the BIS and BAS scales were used as indices of self-reported sensitivity to reward and punishment, respectively.

Procedure

Orthogonalized go/no-go task. The orthogonalized go/no-go task was adapted from a probabilistic reinforcement learning paradigm (Cavanagh, Eisenberg, Guitart-Masip, Huys, & Frank, 2013; Guitart-Masip et al., 2012). This task included two orthogonal factors: valence (reward or punishment) and action (go or no-go), which formed four conditions: 1) go to win reward (GW), 2) no-go to win reward (NGW), 3) go to avoid punishment (GA), and 4) no-go to avoid punishment (NGA). On each trial (Fig. 1A), participants were presented with one of the four cues, each of which predicted a unique combination of the optimal action (go or no-go) and potential outcome (reward or punishment). Participants were not explicitly told the action contingencies for each cue. Thus, they need to explore both actions (i.e., respond to the target or withhold a response) to learn how to achieve the best outcome for each cue. Since the valence-action associations in the GW and NGA conditions were consistent with Pavlovian pre-programmed behavioral tendency, these two conditions were Pavlovian-congruent. In contrast, the GA and NGW conditions were Pavlovian-incongruent (Kim et al., 2023).

As shown in Fig. 1A, each trial began with the display of a cue followed by a variable interval. Then a white circle as target appeared in the middle of the screen that required the participants to respond (go or no-go). After the offset of the circle, the feedback was presented indicating a reward outcome (green +$), a punishment outcome (red -$), or a neutral outcome (a yellow bar indicating no reward or punishment). A fixation cross with a variable interval was presented as the intertrial interval. Similar to many other RL tasks, the feedback of the current task was probabilistic. Specifically, on reward trials, 75% of correct responses and 25% of incorrect responses were rewarded (and the remaining 25% of correct and 75% of incorrect responses led to the neutral outcome). On punishment trials, 75% of correct responses and 25% of incorrect responses avoided losing money (Fig. 1B).

The task included three blocks, each with fifteen trials per experimental condition (GW, NGW, GA, NGA). Therefore, there were a total of 180 trials with 45 trials for each condition. The trials were randomly presented within each block. A practice session was conducted before the formal task to help participants to get familiar with the task.

Behavioral data analysis

The behavioral data were analyzed using R (version 4.3.2; https://cran.r-project.org/). Between-group comparisons of demographic and clinical characteristics were conducted using independent-sample t-tests or chi-squared tests.

For the task performance, a mixed ANOVA on response accuracy was conducted with valence (reward/punishment) and action (go/no-go) as within-subject factors, and group (PSMU/HC) as a between-subject factor.

Similar to previous studies (Kim et al., 2023), we have also calculated two Pavlovian bias indices (PBI), defined as the difference between the accuracy of Pavlovian-congruent and Pavlovian-incongruent conditions in the reward and punishment domains, respectively:

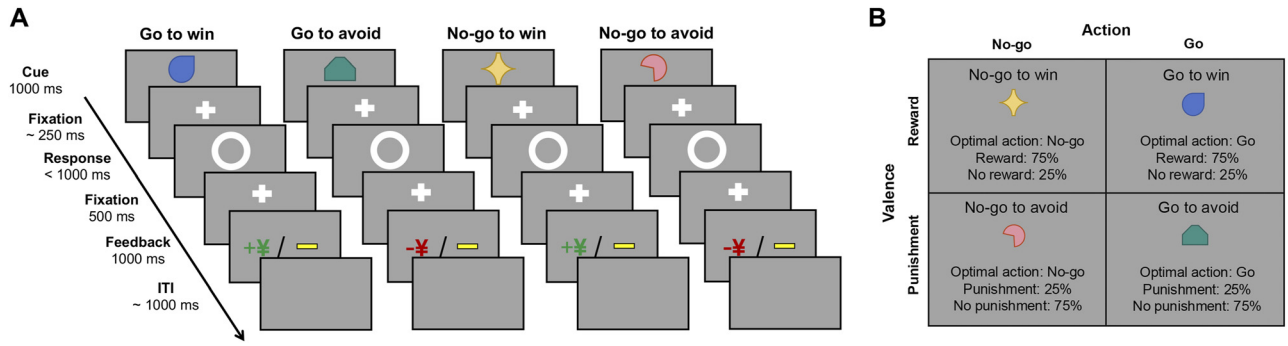

Fig. 1. Task design overview. (A) A schematic illustration of the orthogonalized go/no-go task. Four types of stimuli were presented. Two conditions were Pavlovian-congruent: go action to win reward and no-go action to avoid punishment. The other two conditions were Pavlovian-incongruent: go action to avoid punishment and no-go action to win reward. (B) Valence-action contingencies in the four experimental conditions. The optimal action (go or no-go), possible outcomes, and their probabilities after a correct response to the target (in each cell)

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$$PBI\ (reward) = Accuracy\ (go\ to\ win) - Accuracy\ (nogo\ to\ win)$$

$$PBI\ (punishment) = Accuracy\ (nogo\ to\ avoid) - Accuracy\ (go\ to\ avoid)$$

Computational modeling construction

Five nested reinforcement learning models used in previous research have been applied to examine latent parameters thought to underlie individual differences in behavioral performance (Guitart-Masip et al., 2012). For example, Model 5 (the best model of the current study) assigned each action a associated with a stimulus s on trial t a probability based on a weight function $W_t(a, s)$. A softmax choice function was used to predict the most likely action on each trial (Kim et al., 2023).

$$p(a_1|s) = \left\{\frac{\exp(W_t(a_1, s))}{\exp(W_t(a_1, s)) + \exp(W_t(a_2, s))}\right\}(1 - \xi) + \frac{\xi}{2} \quad (1)$$

Here, the probability of a go action was estimated using a squashed softmax function and based on the weights for go (a_1) and no-go (a_2). The parameter ξ was the irreducible noise ($0 \leq \xi \leq 1$), representing the randomness of action selection. The action weight function $W_t(a, s)$ for the two reward conditions (GW and NGW) was defined as follows:

$$W_t(a, s) = \begin{cases} Q_t(a, s) + b + \pi_{rew}\ V_t(s) & if\ a = go \\ Q_t(a, s) & if\ a = nogo \end{cases} \quad (2)$$

The parameter b was a static bias for go action, which modified the action weight across valence. The parameter π_{rew} was the Pavlovian bias reflecting the behavioral approach to stimuli that had positive learned value. The action weight function $W_t(a, s)$ for the two punishment conditions (GA and NGA) was the same as Eq. (2) except that the parameter π_{rew} was replaced by π_{pun}, another Pavlovian bias parameter that reflecting behavioral inhibition for stimuli with negative learned value.

The Q values were updated according to a Rescorla-Wagner like rule. For the two reward conditions (GW and NGW), a reward sensitivity parameter ρ_{rew} scaled the feedback value r and a learning rate parameter ε scaled the update term, as follows:

$$Q_t(a_t, s_t) = Q_{t-1}(a_t, s_t) + \varepsilon\ (\rho_{rew} r_t - Q_{t-1}(a_t, s_t)) \quad (3)$$

The expected value of each stimulus V_t was updated using a similar rule. V_t was updated according to the difference between current feedback r_t and previous expected value V_{t-1} to the stimulus:

$$V_t(s_t) = V_{t-1}(s_t) + \varepsilon\ (\rho_{rew} r_t - V_{t-1}(s_t)) \quad (4)$$

For the two punishment conditions (GA and NGA), the Q and V values were updated in a similar manner, except that a punishment sensitivity parameter ρ_{pun} scaled the feedback value r:

$$Q_t(a_t, s_t) = Q_{t-1}(a_t, s_t) + \varepsilon\ \left(\rho_{pun} r_t - Q_{t-1}(a_t, s_t)\right) \quad \text{Eq(5)}$$

$$V_t(s_t) = V_{t-1}(s_t) + \varepsilon\ \left(\rho_{pun} r_t - V_{t-1}(s_t)\right) \quad \text{Eq(4)}$$

The other four models are reduced versions of Model 5 (Fig. 3). For example, Model 4 used a single Pavlovian bias parameter π for both the reward and punishment conditions; Model 3 used the same Pavlovian bias π and feedback sensitivity ρ for the reward and punishment conditions; Model 2 did not consider the Pavlovian bias; and Model 1 did not consider the Pavlovian bias and static bias for the go action.

Model parameter estimation

The model parameters were estimated using hierarchical Bayesian analysis (HBA), as it can generate more robust estimates and is less sensitive to outliers compared to other estimation methods, such as maximum likelihood estimation (Gelman & Shalizi, 2013). HBA was conducted based on Markov chain Monte Carlo (MCMC) algorithms as implemented using Stan (version 2.21; https://mc-stan.org/) and hBayesDM package in R (Ahn, Haines, & Zhang, 2017; Gelman & Shalizi, 2013). Similar to the models for the orthogonalized Go/no-go task in the hBayesDM library (Ahn et al., 2017), we used the weakly informative priors for all parameters (Table S2). The models for the PSMU and HC groups were fitted the separately to generate stable and reliable individual estimates (Kim et al., 2023). We used four independent chains and a sample size of 4,000, including 2,000 burn-in samples per chain. The convergence of the chains was confirmed by inspecting the Rhat values (Rhat <1.1) and the trace plots of four chains.

Model comparison

The leave-one-out information criterion (LOOIC) values were used to compare the models regarding their goodness-of-fit (Vehtari, Gelman, & Gabry, 2017). The LOOIC is a Bayesian criterion to evaluate the out-of-sample predictive performance of a model, and a lower LOOIC indicates a better model fit. The model weight, an index of the relative likelihood of a model calculated using Bayesian model averaging, was also calculated (Yao, Vehtari, Simpson, & Gelman, 2018). The loo package (version 2.7.0; https://cran.r-project.org/web/packages/loo/) in R was used to calculate the LOOIC and model weight for each model.

Since model 5 showed best model fitting for both PSMU and HC groups, we conducted the group comparison based on this model. Specifically, the posterior distribution of the HC group was subtracted from that of the PSMU group. Group differences were considered credible when the 90% highest density intervals (HDI) of posterior difference distributions did not include the value 0 (Kreis, Zhang, Moritz, & Pfuhl, 2022; Langley et al., 2023).

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Ethics

The study procedures were carried out in accordance with the Declaration of Helsinki. The Institutional Review Board of the Sun Yat-sen University approved the study. All subjects were informed about the study and all provided informed consent.

RESULTS

Demographic and behavioral results

The PSMU and HC groups were matched for age, gender, and education. As expected, the PSMU group showed significantly higher problematic social media use severity relative to the HC group. The PSMU group seemed to spend more time using social media than the HC group, although the difference was not significant (Table 1).

A mixed ANOVA on accuracy was also conducted with valence (win/lose) and action (go/no-go) as within-subject factors, and group (PSMU/HC) as a between-subject factor. We found that the interaction effect of valence by action by group ($F_{(1,63)} = 3.97$, $p = 0.051$, partial $\eta^2 = 0.06$) was marginal significant. Moreover, the interaction effects of valence by group ($F_{(1,63)} = 4.25$, $p = 0.043$, partial $\eta^2 = 0.06$), and action by group ($F_{(1,63)} = 4.78$, $p = 0.032$, partial $\eta^2 = 0.07$) were both significant. The simple effect analysis showed that the PSMU group performed significantly worse than the HC group in the no-go to win condition ($F_{(1,63)} = 7.66$, $p = 0.007$, partial $\eta^2 = 0.11$; Fig. 2A); however, the between-group difference was insignificant in the other three conditions.

Similar to the ANOVA findings, we found that the PSMU group showed a significantly higher Pavlovian bias index for the reward domain, but not the punishment domain, compared with the HC group (Table 1, Fig. 2B). The learning curves across trials indicated that the two groups performed similarly in most conditions, except for the no-go to win condition (Fig. 2C). Therefore, we turned to a computational modeling analysis to further elucidate the latent processes underlying the condition-specific learning characteristics and between-group differences based on the reinforcement learning theory.

Computational modeling results

Reinforcement learning modeling was proceeded according to previous research (Guitart-Masip et al., 2012; Kim et al., 2023). We used five parameterized reinforcement learning models (Fig. 3A) to fit the behavioral data of the participants, and model comparison was conducted separately for each group. Based on the LOOIC values, Model 5 (with separate outcome sensitivity and Pavlovian bias parameters for the reward and punishment domains) showed the best fitting (lowest LOOIC values) for both groups (Fig. 3A).

Since the learning characteristics of the PSMU and HC groups during the orthogonalized go/no-go task were best captured by the same model, we thus calculated the posterior distributions and performed between-group comparison on all group-level parameters of Model 5. As shown in Fig. 3C, the PSMU group showed higher learning rate (ε) and lower reward sensitivity (ρ_{rew}). No credible between-group differences were found in the other five parameters (Table 2). The posterior distributions of these five group-level parameters can be found in Fig. S1.

The current study focused on the between-group comparison on group-level parameters. Another analytical approach is to treat all participants as a single group and examine the relationship between the parameter values and PSMU severity (i.e., BSNAS scores) across all participants. To examine whether our findings were generally consistent across these two analytical approaches, we also fitted Model 5 to all participants and performed post-hoc correlation analyses between the PSMU severity and model parameter values and model-free measures. We found that the PSMU

Table 1. Demographic information, social media use characteristics, and task performance

Variables	Mean (SD)		t-test/χ^2	p	Effect size
	PSMU (n = 33)	HC (n = 32)			
Age	20.52 (1.60)	21.25 (1.77)	−1.75	0.09	d = 0.43
Gender (f/m)	26/7	20/12	1.37	0.24	V = 0.15
Education	14.91 (1.40)	15.09 (1.33)	−0.55	0.59	d = 0.13
BSMAS	22.91 (2.54)	13.19 (3.21)	13.57	<0.001	d = 3.36
Gaming time (hrs/week)	5.11 (7.45)	2.60 (2.90)	1.79	0.08	d = 0.44
Video and music use (hrs/week)	17.22 (13.37)	13.23 (12.56)	1.24	0.22	d = 0.39
Online social network (hrs/week)	19.32 (18.34)	14.28 (14.98)	1.47	0.21	d = 0.30
Impulsivity scale[a]	72.07 (7.15)	67.42 (7.17)	2.17	0.04	d = 0.65
BIS scale[a]	20.70 (2.80)	19.26 (3.35)	1.59	0.12	d = 0.48
BAS scale[a]	40.33 (4.14)	41.74 (5.58)	−0.98	0.33	d = 0.30
PBI - reward domain	0.50 (0.36)	0.28 (0.41)	2.39	0.02	d = 0.57
PBI - punishment domain	−0.04 (0.22)	−0.02 (0.21)	−0.29	0.77	d = 0.09

BSMAS: Bergen social media addiction scale; BIS: Behavioral Inhibition System scale; BAS: Behavioral Approach System scale; PBI: Pavlovian bias index. PSMU: problematic social media use; HC: healthy control.
[a] A subset of 46 participants (19 HC and 27 PSMU) completed the scales.

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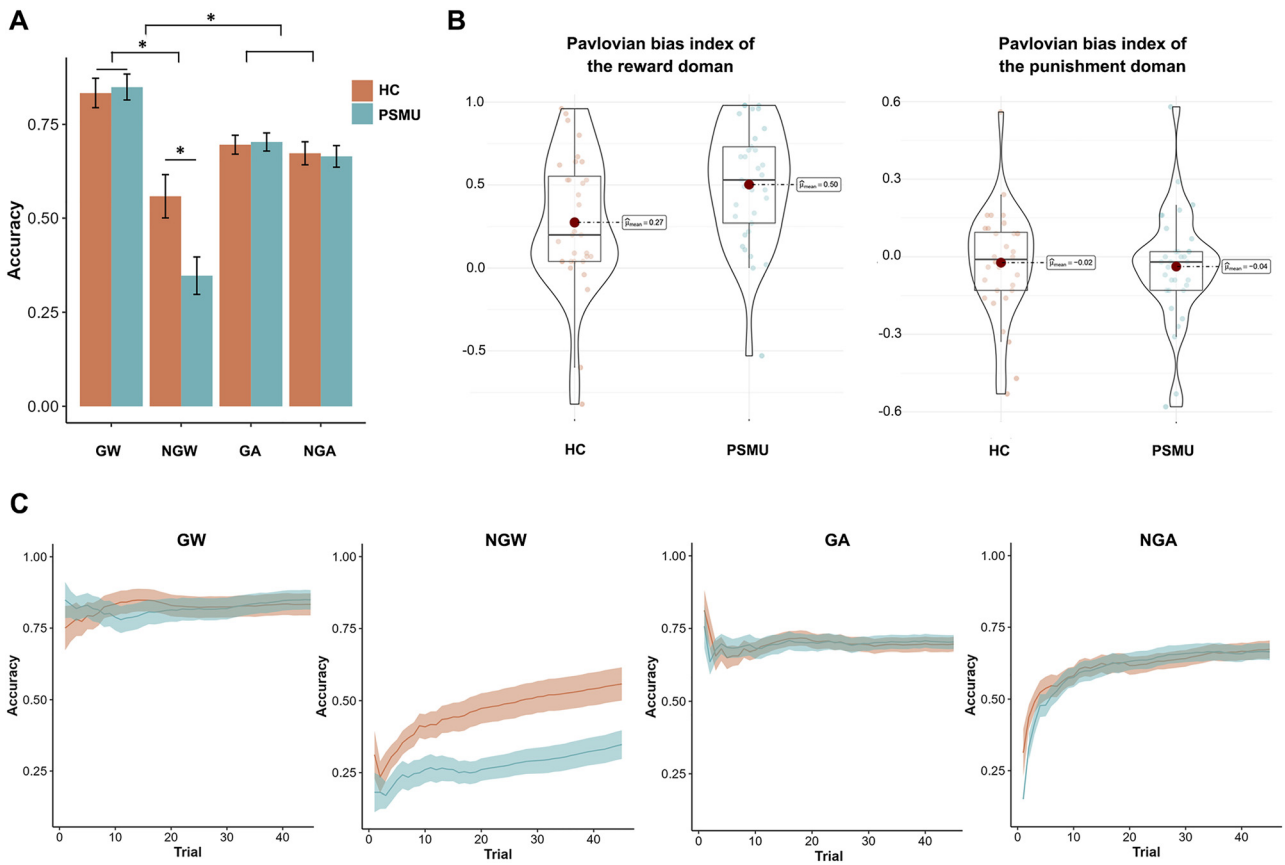

Fig. 2. Model-free results of the orthogonalized go/no-go task. (A) The PSMU group showed decreased accuracy in the no-go to win condition, reflecting a higher Pavlovian bias in the reward domain compared with the HC group. (B) The PSMU group showed a higher behavioral Pavlovian bias index (congruent accuracy - incongruent accuracy) in the reward domain, but not in the punishment domain, compared with the HC group. (C) Learning curves across trials indicated that the two groups performed similarly in most conditions, except for the no-go to win condition.
Error bars indicate standard error of mean

severity was negatively associated with reward sensitivity (Spearman's $\rho_{(63)} = -0.66$, $p < 0.001$) and punishment sensitivity ($\rho_{(63)} = -0.27$, $p = 0.03$). The relationship between PSMU severity and learning rate was insignificant ($\rho_{(63)} = 0.15$, $p = 0.22$). All other associations were negligible (Table S3). For the model-free indices, the PSMU severity was marginally associated with the accuracy of the NGW condition ($\rho_{(63)} = -0.23$, $p = 0.07$; Table S4).

Correlation between computational, behavioral, and self-reported measures

Finally, we explored the relationship between the key parameter values (i.e., learning rate, reward sensitivity, and punishment sensitivity), model-free task performance (i.e., accuracy of the two incongruent conditions), and self-reported measures in all participants. As shown in Table S5, we found that the reward sensitivity parameter values correlated negatively with impulsivity ($\rho_{(44)} = -0.30$, $p = 0.04$) and positively with BAS scores ($\rho_{(44)} = 0.31$, $p = 0.04$). The associations between the accuracy of the two incongruent conditions and the self-reported measures were not significant (Table S6). Moreover, we found that the reward sensitivity was marginally correlated with the accuracy of the NGW condition ($\rho_{(63)} = 0.24$, $p = 0.05$). The learning rate was positively associated with the accuracy of the NGW condition ($\rho_{(63)} = 0.26$, $p = 0.04$) and the GA condition ($\rho_{(63)} = 0.43$, $p < 0.001$), although these effects were mainly driven by the HC group ($\rho_{(30)} = 0.51$, $p = 0.003$ for NGW and $\rho_{(30)} = 0.46$, $p = 0.008$ for GA) instead of the PSMU group ($\rho_{(31)} = 0.12$, $p = 0.49$ for NGW and $\rho_{(31)} = 0.39$, $p = 0.03$ for GA). More details on the associations between the key parameter values and task performance can be found in Table S6.

DISCUSSION

This study used a combination of an orthogonalized go/no-go task and computational modeling to examine how individuals with PSMU were influenced by Pavlovian bias during instrumental learning. We found that: 1) the PSMU group exhibited a higher Pavlovian bias index in the reward domain than the HC group; 2) the computational modeling results showed an increased learning rate and a decreased reward sensitivity in the PSMU group than the HC group.

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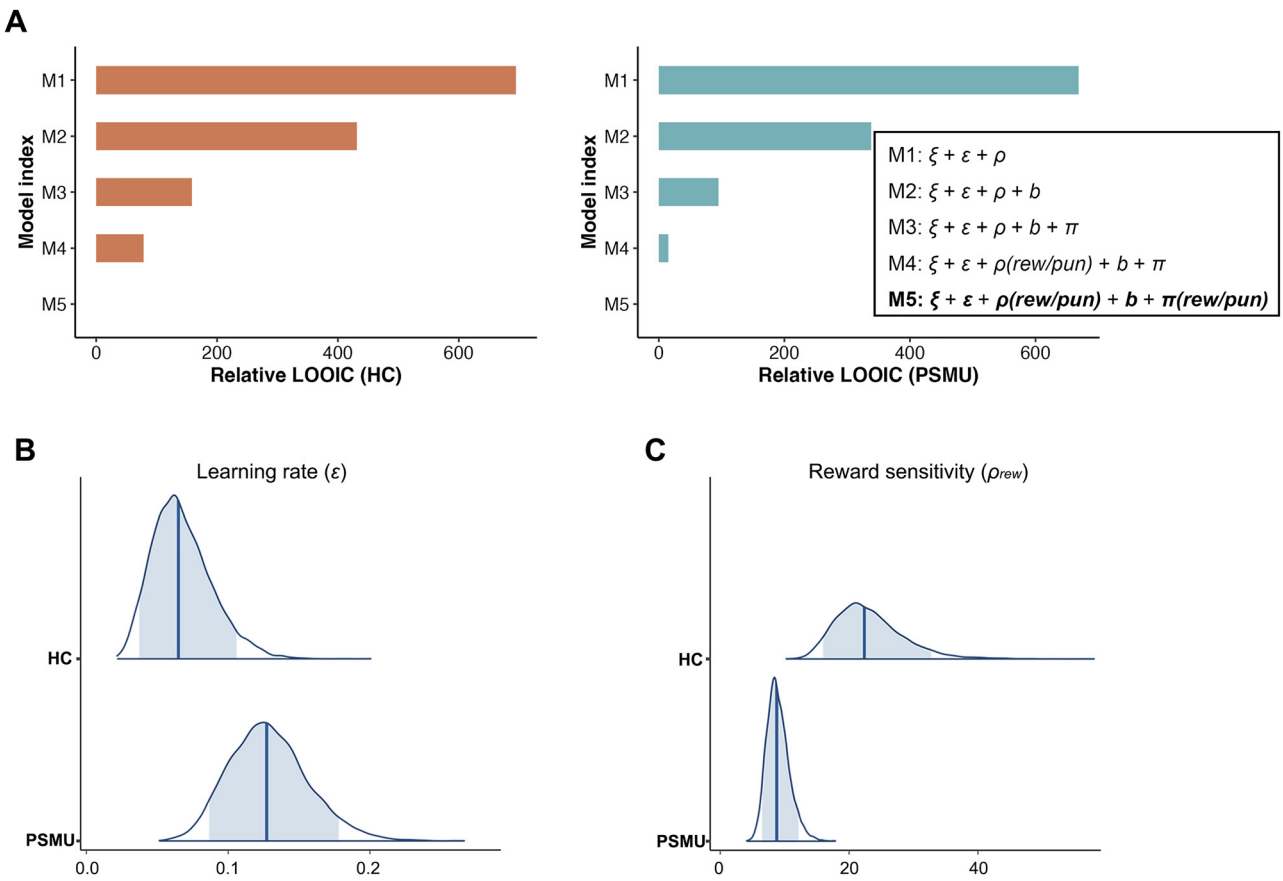

Fig. 3. Computational modeling results of the orthogonalized go/no-go task. (A) Model comparison showed that Model 5 was the best for both the HC and PSMU groups. Lower LOOIC values indicated better model performance. ξ: irreducible noise; ε: learning rate; ρ: sensitivity to outcomes; ρ_{rew}: sensitivity to reward outcomes; ρ_{pun}: sensitivity to punishment outcomes; b: go bias; π: Pavlovian bias. π_{rew}: Pavlovian bias in the reward domain; π_{pun}: Pavlovian bias in the punishment domain. (B–C) The PSMU group showed a higher learning rate and lower reward sensitivity compared with the HC group. Posterior distributions of the group-level parameters were taken from Model 5

Table 2. Posterior mean (90% highest density interval) of group-level parameters in the PSMU and HC groups

Parameters	PSMU	HC	Difference
Irreducible noise (ξ)	0.065 [0.041, 0.094]	0.079 [0.051, 0.115]	−0.016 [−0.057, 0.026]
Learning rate (ε)	**0.127 [0.087, 0.178]**	**0.065 [0.037, 0.106]**	**0.061 [0.003, 0.118]**
Go bias (b)	0.894 [0.697, 1.110]	0.854 [0.620, 1.120]	0.037 [−0.300, 0.345]
Pavlovian bias - reward (π_{rew})	0.202 [0.039, 0.401]	0.053 [−0.230, 0.362]	0.151 [−0.202, 0.494]
Pavlovian bias - punishment (π_{pun})	0.431 [0.268, 0.595]	0.428 [0.234, 0.636]	0.002 [−0.272, 0.243]
Sensitivity - reward (ρ_{rew})	**8.770 [6.490, 12.200]**	**22.400 [16.000, 32.800]**	**−14.170 [−22.756, −5.182]**
Sensitivity - punishment (ρ_{pun})	5.460 [4.230, 7.100]	8.380 [6.160, 11.800]	−3.077 [−6.209, 0.070]

The credible differences between the PSMU and HC groups are indicated in bold font.

Taken together, the study provides new insights into how Pavlovian bias influences instrumental learning and the underlying computational mechanisms in individuals with PSMU.

In the orthogonalized go/no-go task, two interdependent axes of behavioral control, valence and action, were independently manipulated, which allowed us to comprehensively examine the interaction between Pavlovian and instrumental systems during associative learning when they are congruent or in conflict (Guitart-Masip et al., 2014). Our findings suggest that the instrumental learning in the reward domain was overly impacted by the Pavlovian pre-programmed behavioral tendency in the PSMU group, in parallel with studies using the PIT task that found enhanced specific PIT effects in IGD participants (Qin et al., 2023; Vogel et al., 2018; Xu et al., 2024). For example, a recent study revealed that individuals with IGD showed fewer button presses to cues with negative values and more button presses to cues with positive values in Go trials (Xu et al., 2024). Although the evidence based on the same paradigm is still needed, these findings suggest that individuals with different types of problematic usage of the internet (e.g., PSMU and IGD) may overly rely on the Pavlovian system, which may interfere with the use of flexible instrumental learning.

Furthermore, previous studies in SUDs have also shown a higher Pavlovian bias or habitual learning in general at the

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expense of goal-directed instrumental learning (Ersche et al., 2016; Sebold et al., 2014). For example, a stronger PIT effect has been reported to be closely associated with alcohol use disorder severity (Garbusow et al., 2022), and the behavioral deficits may be linked to a decreased lateral prefrontal cortex activity and an increased ventral striatum activity (Chen et al., 2021). Taken together, these findings suggest that an over-reliance of Pavlovian system during associative learning seems to be a key characteristic of addiction in general (Chen et al., 2021; Garbusow et al., 2022). However, it should be noted that demographic (e.g., age and medication status) and methodological (e.g., paradigms and analyses) differences between SUD and PSMU studies warrant careful consideration. Direct comparisons based on matched samples and the same paradigm are still needed for future research.

To more precisely investigate latent factors that may contribute to associative learning during the orthogonalized go/no-go task, we tested five nested computational models (Guitart-Masip et al., 2012; Kim et al., 2023). The model comparison results indicated that the same model showed the best fit for both the PSMU and HC groups. Except for common parameters, such as learning rate and static action bias, this model also distinguished between outcome sensitivity and Pavlovian bias parameters related to reward and punishment. Altered reward processing has been highlighted as a central feature of different types of addictive disorders (Brand et al., 2019; Dong & Potenza, 2014; Li et al., 2020; Yao et al., 2017, 2022). The decreased reward sensitivity and blunted reward prediction error signals in the reward system at the neural level have been reported in SUDs and IGD (Lei et al., 2022; Rose et al., 2014; Tanabe et al., 2013). Our study extends this important question to the context of PSMU and suggests that insensitivity to monetary rewards may play a critical role in associative learning deficits in PSMU.

In addition to reward sensitivity, our modeling results also showed an increased learning rate in the PSMU group. Learning rate is critical for goal-directed behavior because it needs to be adapted to environmental changes (O'Reilly, 2013). As a computational measure, the goodness of learning can be considered as an index of how well people adapt their learning rate to the changing environment (Lockwood et al., 2020; Scholl & Klein-Flügge, 2018). Importantly, another recent reinforcement learning study in other addictions (i.e., IGD and SUDs) also identified an increased learning rate (Kwon, Choi, Park, Ahn, & Jung, 2024). It should be noted, however, that the optimal learning rate depends on the nature of the task and may vary between studies (Lockwood et al., 2020). According to the winning model of the current study, the reward sensitivity and learning rate together control learning from the previous feedback (Eq. (3)). Therefore, one possibility is that the PSMU group increased the learning rate to counteract the negative effects of the blunted reward sensitivity on feedback learning. However, such compensation may be ineffective in situations that require more flexible behavior (e.g., the NGW condition).

While researchers have devoted much effort to applying theoretical models of addiction to PSMU, few empirical studies have directly tested whether the hypotheses from these theories still hold in PSMU (Wegmann et al., 2020). The current study partially addressed this research gap by examining several critical factors included in the theoretical models of addiction. For example, in line with the influential I-PACE model (Brand et al., 2019; Wegmann et al., 2020), we found that individuals with PSMU showed poorer performance in a task that requires the interaction of several affective, cognitive, and executive processes (e.g., reward expectation, decision-making, and feedback learning). Beyond the framework of the original I-PACE model, our study further elucidated how some latent factors, including reward sensitivity and learning rate, interact with each other using a computational modeling approach. Moreover, the poor performance of the PSMU group in the NGW condition may not be due to a general impairment of inhibitory control, since they showed similar performance in the GA and NGA conditions and three bias values from the computational model compared with the HC group. However, it should be noted that this evidence is rather indirect, as this task is not designed to specifically measure inhibitory control. Therefore, a promising future direction is to incorporate multiple paradigms to investigate whether and how the general or PSMU-specific inhibitory control may influence the decision-making and learning, as suggested by the I-PACE model (Brand et al., 2019).

The findings of the current study may also shed new light on the development of tailored interventions for PSMU. For example, the motor response training (also based on a go/no-go paradigm) may be used to reduce the impact of stimulus-outcome associations preprogrammed in the Pavlovian system on instrumental learning (Chen, Veling, Dijksterhuis, & Holland, 2016), although its efficacy in alleviating PSMU symptoms has not been tested yet. In addition, neuromodulatory methods, such as non-invasive brain stimulation or real-time neurofeedback, may be used to change instrumental learning and related cognitive processes (Kahnt, 2023). For example, anodal transcranial direct current stimulation (tDCS) over the dorsolateral prefrontal cortex has been shown to reduce Pavlovian bias in the orthogonalized go/no-go task (Kim et al., 2023). However, the effect seems to be stronger in the punishment domain, but not in the reward domain. Therefore, future studies should further explore interventions that can reduce reward-related Pavlovian bias and PSMU severity.

Despite the above advantages, this study has some limitations. First, we only assessed the learning performance at the behavioral level. The interaction between valence and action has been suggested to be under the modulation of the dopaminergic system (Guitart-Masip et al., 2012). Future work to link these findings to potential neuromodulatory measures is needed. Second, although the sample size of the current study is similar to many other computational modeling studies (Hoven, Luigjes, & van Holst, 2024; Kwon et al., 2024; Wiehler, Chakroun, & Peters, 2021) and meets the basic requirement for most parametric analyses, future studies with large sample sizes are still needed to confirm the findings of this study. Particularly, only a subset of participants completed the questionnaires, so the related correlation

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results should be considered preliminary. Third, we did not find between-group differences related to reinforcement learning in the punishment domain, possibly because the monetary losses used in our task did not induce strong punishment effects. Studies using more effective punishment manipulation (e.g., an electric shock) are still needed to investigate the influence of Pavlovian bias on instrumental learning in the punishment domain in PSMU. Moreover, we did not include other cognitive tasks in the current study. It would be interesting to examine the relationship between performance on the orthogonalized go/no-go task (including modeling results) and basic executive functions such as working memory and general inhibitory control in individuals with PSMU. Finally, this study used young adults with high severity of PSMU. Future studies are encouraged to test if our findings could be generalized to clinical samples.

CONCLUSIONS

In summary, this study systematically investigated the impact of Pavlovian pre-programmed behavioral tendencies on instrumental learning in PSMU by dissociating the roles of valence and action. Individuals with PSMU showed worse learning performance, especially when the Pavlovian and instrumental systems were in conflict. Computational modeling further showed an increased learning rate and a blunted reward sensitivity in the PSMU group. These findings advance our understanding of how Pavlovian pre-programmed behavioral tendencies bias instrumental learning in PSMU, shedding new light on extending classical addiction theory to the field of PSMU and developing more effective interventions for this disorder.

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Authors' contribution: LL and Y-WY designed the study. LL, Y-XP, Z-HS, S-JC, and Y-YH collected the data. LL and Y-WY performed analysis. LL administrated the project. LL and Y-WY wrote the manuscript with the input from all the other authors.

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SUPPLEMENTARY MATERIAL

Supplementary data to this article can be found online at https://doi.org/10.1556/2006.2025.00026.

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